Quality Control Analysis Report

TST Jurisprudence Pipeline - CSDS-352 Capstone

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1 Quality Control Analysis Report

1.1 1. Scope and objective

This report documents the quality control (QC) module of the TST jurisprudence pipeline. It describes how candidate quality problems are detected, presents the distributions and scores produced over the full corpus of 1058 verbetes, and discusses the concrete anomalies that the module flags for manual review.

The module answers two complementary questions:

1. **Is this record structurally complete?** (heuristic rules + supervised classifier)
2. **Is this record unusual with respect to the rest of the corpus?** (unsupervised anomaly detection)

1.2 2. Method

1.2.1 2.1 Heuristic labelling

During ingestion each verbete receives a heuristic `quality_label`:

Label	Rule
`cancelado`	the text contains a cancellation marker (e.g. “cancelada”)
`curto`	text length below `VERBETE_CURTO` (120 characters)
`ruido`	text length below `VERBETE_RUIDO` (30 characters)
`duplicado`	the body is repeated under more than one code
`completo`	none of the above

1.2.2 2.2 Supervised re-labelling

A TF-IDF vectoriser (5000 features, 1-2 grams) feeds a Logistic Regression (`class_weight="balanced"`) trained on the heuristic labels with an 80/20 stratified split. The fitted classifier then predicts the quality label for every verbete, and the trained vectoriser and classifier are persisted to `models/quality_classifier.joblib`.

1.2.3 2.3 Anomaly detection

An IsolationForest (200 trees, 5% contamination) is fitted on the 384-dimensional sentence embeddings. Its decision function is rescaled to a 0-1 `anomaly_score` (higher is more anomalous) and stored with each verbete.

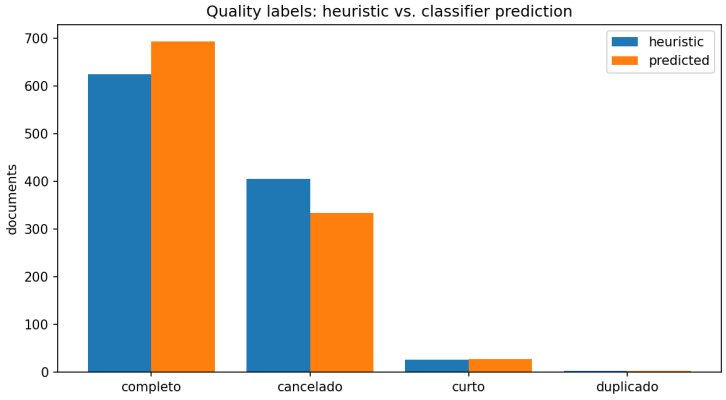
1.2.4 2.4 How to read the metrics

- **Precision** - of the records the classifier *predicted* as a given class, the fraction that truly belong to it; it measures the trustworthiness of a positive prediction.
- **Recall** - of the records that *actually* belong to a class, the fraction the classifier found; it measures coverage.
- **F1** - the harmonic mean of precision and recall, a single balanced number.
- **Accuracy** - the fraction of all predictions that are correct; it can be misleading when classes are imbalanced, which is why F1 is also reported.
- **Macro F1** - the average F1 over classes, weighting every class equally, so a tiny class counts as much as a large one; it is low here because `curto` scores zero.
- **Weighted F1** - the average F1 weighted by class size, so it is dominated by the large classes. The gap between macro and weighted F1 quantifies how much the tiny classes hurt the average.
- **Anomaly score** - the IsolationForest output rescaled to 0-1, where higher means more unusual relative to the rest of the corpus. It is a *review cue*, not a judgment about legal validity.

1.2.5 2.5 Why these methods

- **TF-IDF + logistic regression** was chosen because it is fast, interpretable (you can inspect the learned word weights) and strong on this scale of data; a heavier model would be harder to justify with heuristic labels.
- **IsolationForest** is unsupervised, so it can flag novelty without any labels, complementing the supervised classifier.
- **UMAP + HDBSCAN** were chosen over PCA + k-means because UMAP preserves local neighbourhoods for visualisation, and HDBSCAN needs no preset number of clusters and marks outliers as noise.

1.3 3. Quality label distribution



Heuristic versus predicted quality labels

Label	Heuristic	Predicted
completo	625	694
cancelado	405	334
curto	26	27
duplicado	2	3
Total	**1058**	**1058**

The classifier moves 71 documents from `cancelado` to `completo` and reassigns a handful of short entries, reflecting the fact that the lexical cancellation marker is not always a reliable indicator of an actually cancelled record.

1.4 4. Classifier evaluation

Held-out test split: 212 documents (846 for training).

Metric	Value
Accuracy	0.906
F1 (macro)	0.708
F1 (weighted)	0.896

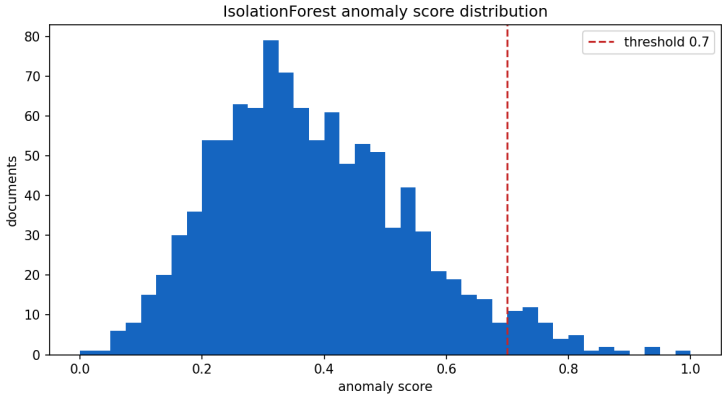
Per-class results:

Class	Precision	Recall	F1	Support
cancelado	0.99	0.84	0.91	81
completo	0.87	0.98	0.92	125

Class	Precision	Recall	F1	Support
curto	0.00	0.00	0.00	5
duplicado	1.00	1.00	1.00	1

The two dominant classes (`completo`, `cancelado`) are classified reliably. The `curto` class is too small to be learned and is always missed, which drags the macro F1 down; in practice short records are still surfaced by the length rule at ingestion and by the anomaly score.

1.5 5. Anomaly detection



IsolationForest anomaly score distribution

- Mean anomaly score: 0.382.
- Documents above the 0.7 review threshold: **47 (4.4% of the corpus)**.

The top flagged records concentrate on historical Precedentes Normativos about rural work, whose vocabulary differs sharply from the modern Súmulas and OJs. The ten highest-scoring documents are:

doc_id	tipo	orgao	quality	score
precedente_106	precedente	SDC	completo	1.000
precedente_63	precedente	SDC	completo	0.949
precedente_107	precedente	SDC	completo	0.936
precedente_60	precedente	SDC	completo	0.896
oj_sbdi-1_252	oj	SBDI-1	cancelado	0.864
precedente_115	precedente	SDC	completo	0.852
sumula_83	sumula	TST	cancelado	0.829
precedente_22	precedente	SDC	completo	0.824

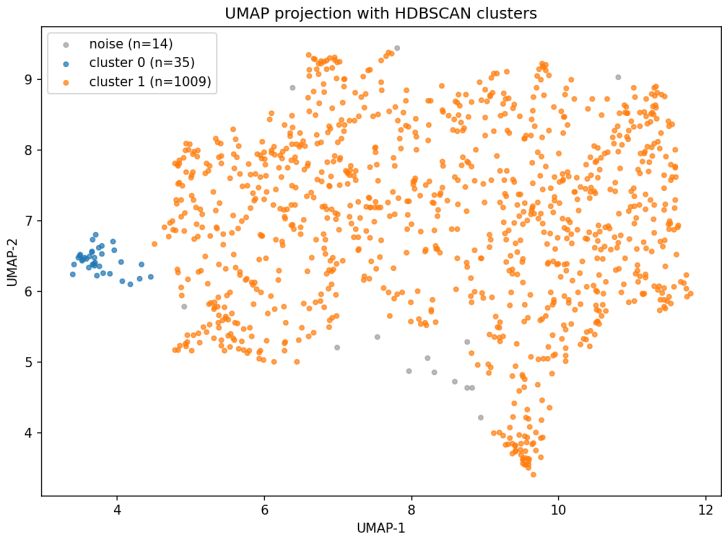
doc_id	tipo	orgao	quality	score
precedente_62	precedente	SDC	completo	0.822
precedente_65	precedente	SDC	curto	0.816

Two representative flagged texts:

- `precedente_106` (score 1.000): “Os empregadores que se dedicarem à pecuária leiteira fornecerão, diariamente, 1 (um) litro de leite aos trabalhadores que exerçam atividades insalubres.”
- `oj_sbdi-1_252` (score 0.864, `cancelado`): “DJ 20, 22 e 25.04.2005 O conceito de mesmo localidade de que trata o art. 461 da CLT...”

The module therefore satisfies the requirement of flagging at least one anomalous or low-quality document, and in fact surfaces a coherent set of them.

1.6 6. Clustering quality



UMAP projection with HDBSCAN clusters

The corpus splits into two dense clusters plus a small noise band:

Cluster	Size	Top terms
1	1009	art., res., para, decorrência, súmula
0	35	concede, homologação, res., sessão
noise	14	-

Cluster 0 captures the historical rural-work precedentes, which is consistent with the anomaly findings in section 5: the same records are both spatially isolated and statistically unusual.

1.7 7. Findings and recommendations

1. **Cancelled records are the largest quality issue** (334 predicted, about 32% of the corpus). They should be filtered from any downstream legal search unless the user explicitly requests them.
2. **The `curto` class needs either more examples or a rule-based fallback**; the supervised classifier cannot learn it from five samples.
3. **Anomalies are mostly genuine and interpretable**, not noise: they are historical precedentes with a distinct legal vocabulary. They are good candidates for a dedicated "historical" collection rather than deletion.
4. **The noise band (14 points)** should be inspected manually; it may contain parsing artefacts.

1.8 8. Reproduction

```
cd capstone
make pipeline                 # produces the labels and scores
python scripts/make_report_figures.py   # regenerates the figures
        and summary.json
```

All numbers in this report are derived from `docs/figures/summary.json`, which the figure script writes from the pipeline artifacts.

1.9 9. Key methods and their sources (in one line each)

1. **Sentence-BERT** (Reimers & Gurevych, 2019) - how to train transformer networks to produce high-quality sentence embeddings; the model family behind our 384-D vectors.
2. **UMAP** (McInnes et al., 2018) - the dimensionality-reduction method used for the 2D projection.
3. **HDBSCAN** (Campello et al., 2013) - the density-based clustering method used to group the corpus and mark noise.
4. **Isolation Forest** (Liu, Ting & Zhou, 2008) - the anomaly-detection method used to score unusual verbetes.

You do not need to read these papers to follow this report; each is summarised above. Full citations appear in the technical report (`report.md`).